Bill Gates, the philanthropist, released mosquitoes into the air at TED 2009 as he was making a point about Malaria.

Siftables

Siftables are technology blocks from MIT, than can be put together like Lego bricks to make different divices. .

hardware has to be reachable by the user; the device has to be as large as the interactive surface; and they all need custom screens or goggles that display the projected images.

All these factors turn portability and cost into insurmountable problems.

Pranav needed to make his gizmo different in several significant ways. First, allow the user to interact with the projected interface using hand gestures. Second, free the user of the limitations of physical space (giant screens) as well as personal (goggles), and make it portable and cost-effective at the same time.

Maes didn't repeatedly praise him as a genius for no reason. A bit of brainstorming, and voila! The SixthSense was born...

So what is sixth sense?

"Other than letting many of you live out your fantasies as Tom Cruise of *Minority Report*, this is one of those devices that can really provide a sixth sense about everything in front of you," Maes said during her address at TED (available on the DVD).

SixthSense is a wearable, gesture interface. The current prototype consists of three main hardware components: a pocket projector (3M MPro110), a camera (Logitech QuickCam) and a laptop computer.

For the software, Pranav developed several computer-vision based techniques/algorithms, using open libraries such as openCV.

The entire cost, sans the laptop computer, runs to about $350 (Rs. 17,500 approximately).

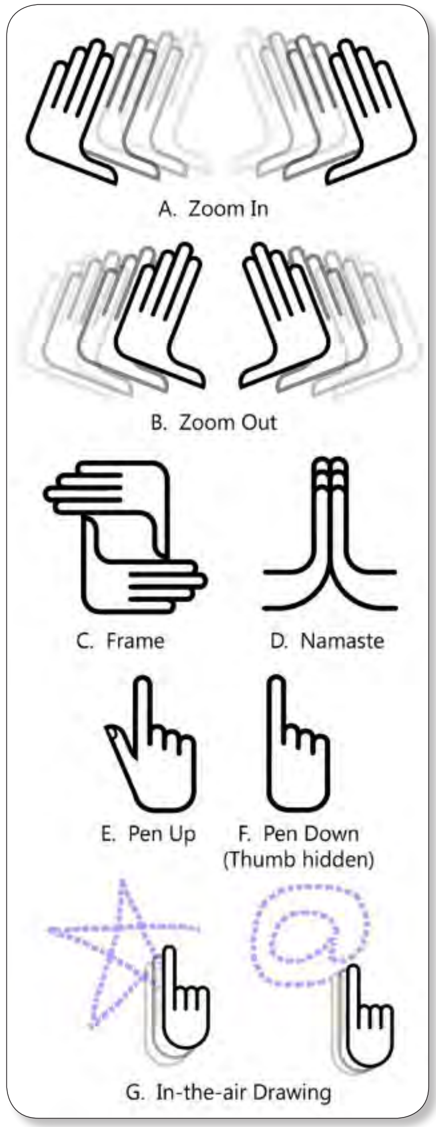
The engineers are discounting the expenses of the laptop because they have already figured out a way to use mobile phones in its place. While the notebook is still used for heavier computation, with mobile devices today sporting the kind of power that a computer had five years ago, it's only a matter of time before laptop is completely replaced by a cellphone.

Both the projector and camera are mounted on a hat or coupled in a pendant-like wearable device, and are connected to the laptop, which the user carries in a backpack.

Pranav prefers the pendant look: "It improves recognition as the body of the user moves less than the head."

"Plus, it would be socially more acceptable in this form factor," he laughs.

The software program processes the video stream data captured by the camera and tracks the locations of the colour markers at the tip of the user's fingers, using simple computer-vision techniques. Currently, these markers are simple caps of whiteboard pens.



The various hand gestures used with SixthSense

"The current prototype only tracks four fingers — two index fingers and two thumbs. This is largely due to the relative importance of the index finger and thumb in natural hand gestures," Pranav explains. Needless to say, the number is currently restricted only by the programming, and could easily be scaled up.

The system also implements a few object-augmentation applications, where the camera recognises and tracks physical objects such as a newspaper, a coffee cup, or a book, and the projector portrays relevant visual information on them.

In a demonstration video (available on the DVD), Pranav walks into a book store and picks up a novel. The Wear Ur World system recognises the book, and portrays its **Amazon.com** rating. It can even provide reviews, user comments, and even additional annotations.

A bit of stylus input...

When it comes to operating an interface based on touch/hand gestures, the first technology to use it effectively was the stylus on touch-screen mobile devices. There's a simple, yet lasting, joy in scribbling a few words on a screen and seeing them digitised. The SixthSense, of course, had to have a pen-type input. The index finger, Pranav thought, would make a good replacement.

"The user can draw on any surface using the movement of the index finger as a pen," he says. Putting the thumb out activates the pen, and putting the thumb back in holsters it.

Of course, this opens up a whole bunch of gesture-based shortcuts: "SixthSense lets the user draw icons or symbols in the air, using the movement of the index finger, and recognises those symbols as interaction instructions."

For example, to access your email, all you have to do is draw the "@" sign in the air, and you'll land up on your inbox. Need to use Google maps? Just draw a star to start plotting your way.

And to undo any action or operation, simply draw an X in the air — the symbol for 'incorrect' doesn't owe its origin to technology, but it is universally recognised!

Still, this is just the tip of the iceberg as far as the SixthSense's gesture-recognition system is concerned.

...a Bit Of The iPhone...

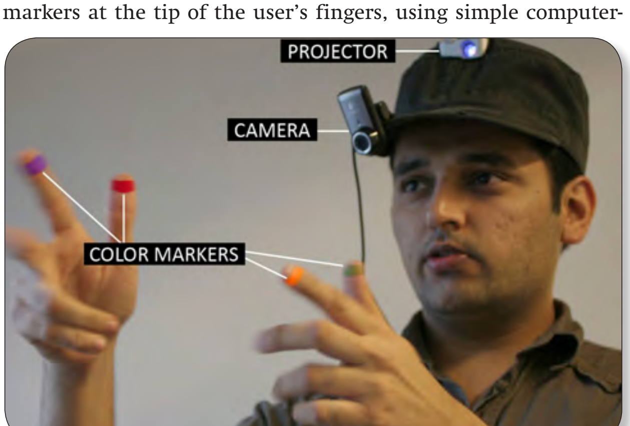
The world of gestures was revolutionised by Apple's iPhone. Not many tech-enthusiasts can forget the moment when Steve Jobs leapt up on stage during his keynote address at Macworld 2007, and announced: "Who wants a stylus? Yuck! We're going to use the best pointing device in the world — our fingers!"

With the popularity that the phone and its different touch-gestures have seen, it just makes sense for a gesture-based device to incorporate these signs.

"SixthSense recognises gestures supported and made popular by interactive multi-touch based products such as Microsoft's Surface or Apple's iPhone," Pranav emphasises.

The gadget's operations are not exactly the same as those for the iPhone, but they are influenced by Apple's technology.

For example, to zoom in on the iPhone, the user pinches



A camera, mini projector and four colour markers later, you are ready to go